Year 13 Maths - Similar to November 2020 Question 2

Similar questions - Pure

Two new questions on the same topic as the exam question, a little more open-ended. They are not past exam questions. Answers are at the end.

1 Two numbers are given by $p = 2^{3a+1}$ and $q = 4^{a+2}$.

(a) Express q in the form 2^k, where k is in terms of a. **[2]**

(b) Show that $\frac{p}{q} = 2^{a-3}$. **[2]**

(c) Given that $\frac{p}{q} = 32$, find the value of a. **[2]**

2 **In this question you must show detailed reasoning.**

(a) Show that $(\sqrt{3}+1)^4 = 28 + 16\sqrt{3}$. **[3]**

(b) Hence write down the value of $(\sqrt{3}-1)^4$ in the form $c + d\sqrt{3}$. **[1]**

(c) Hence show that $(\sqrt{3}+1)^4 \times (\sqrt{3}-1)^4 = 16$. **[2]**

Question 1

(a) $q = (2^2)^{a+2} = 2^{2a+4}$

(b) $\dfrac{p}{q} = 2^{3a+1-(2a+4)} = 2^{a-3}$

(c) $2^{a-3} = 32 = 2^5 \Rightarrow a-3 = 5 \Rightarrow a = 8$

Question 2

(a) $(\sqrt{3}+1)^2 = 4+2\sqrt{3}$, so $(\sqrt{3}+1)^4 = (4+2\sqrt{3})^2 = 16+16\sqrt{3}+12 = 28+16\sqrt{3}$

(b) Replacing $\sqrt{3}$ by $-\sqrt{3}$: $(\sqrt{3}-1)^4 = (1-\sqrt{3})^4 = 28-16\sqrt{3}$

(c) $(28+16\sqrt{3})(28-16\sqrt{3}) = 784-256\times 3 = 784-768 = 16$